

A Machine Learning Approach for FDS Based Power Transformer Moisture Estimation

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Abstract- This paper presents a machine learning approach for estimation of moisture content (MC) in power transformer insulation based on its frequency domain dielectric spectroscopy (FDS). The proposed method is based on principal component analysis and support vector machine approaches. It classifies transformer insulation into three classes based on the estimated MC levels i.e. dry (MC<1%), intermediate (1%<MC<3%) and wet (MC>3%). The proposed algorithm was trained and tested using FDS data sets consisting of 112 laboratory modeled samples and 20 field transformer samples. The laboratory modeled data were estimated using XY model from FDS measurements on pressboard samples having different moisture levels. A range of transformer geometries, aging conditions of pressboard and oil conductivities were considered when deriving the data set. The results showed that the moisture classification was accurate up to 96% thus indicating as a simple and promising method for transformer moisture estimation.

I. INTRODUCTION

Power transformers are considered as one of the critical assets in a power system. Thus, transformer failures may cause power outages, economic losses, and personal and environmental hazards. Insulation degradation due to various electrical, thermal and mechanical stresses is one of the reasons for premature transformer failures [1,2]. In case of oil-paper insulation, presence of moisture significantly accelerates the degradation of the insulation. On the other hand, moisture produced by degradation of paper insulation may easily get diffuse into the oil and hence negatively effect on insulation performance of oil [3,4,5]. Therefore, estimation of moisture content (MC) in a transformer insulation is crucial to avoid unexpected transformer failures.

At present various condition assessment techniques such as dissolved gas analysis (DGA), sweep frequency response analysis (SFRA) and FDS (frequency domain dielectric spectroscopy) are in use for transformer applications. Compared with other techniques, FDS is more sensitive to MC, especially at low frequencies (up to 0.1 Hz) [6-7]. However, FDS is also sensitive to other factors such as temperature, insulation geometry, oil conductivity, and presence of aging by-products, etc. [6-7]. Therefore, it is difficult to establish a simple correlation between FDS curves and the MC levels. Also, current interpretation schemes of FDS diagnosis on transformer insulation may vary due to different database sets used in so called X-Y curve fitting method [8]. A machine learning algorithm can especially be trained to establish correlation between FDS data and the conditions of the corresponding

transformers [8]. The trained algorithm can be used to estimate MC levels in the transformers based on FDS measurements performed periodically.

In the present work, principal component analysis (PCA) method has been proposed to classify the transformer insulation into three categories according to the MC level, such as wet, intermediate, and dry. The method was based on extracting the features of FDS loss tangent curves and identification of principle components (PCs). The first two PCs allow to classify transformers regardless of the aging status, oil conductivity, and insulation geometries. Training and testing of the proposed method were conducted using FDS data sets formulated using field measurements on real transformers as well as laboratory measurements on pressboard samples. Findings of this study confirmed that the proposed model provides an accurate estimation of moisture in the oil-paper insulation of power transformers.

II. THEORY

A. Dielectric response of transformer insulation

Conduction and polarization processes activated due to sinusoidal electric fields leads to power losses in dielectrics. These losses are characterized by so called loss tangent or dissipation factor defined as ratio between real and imaginary capacitances (C' , C'') or permittivity (ϵ' , ϵ'') obtained by frequency response measurements.

$$\tan \delta(\omega) = \frac{C''(\omega)}{C'(\omega)} = \frac{\epsilon''(\omega)}{\epsilon'(\omega)} \quad (1)$$

As far as the transformer insulation is concerned, the X-Y model depicted in Fig.1 can be used to model the dielectric response of composite insulation systems based on the response of individual insulation materials and their geometric details [4]. X and Y parameters defined in equations (2) and (3) can be calculated by using dimensional information provided by transformer manufacturers.

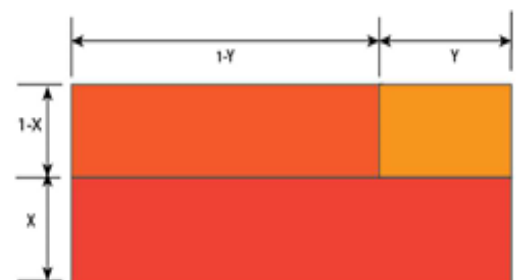


Fig. 1. Transformer XY Model

$$X = \frac{\text{Radial thickness of total barriers}}{\text{Radial width of the duct}} \quad (2)$$

$$Y = \frac{\text{Total width of space along the periphery of the duct}}{\text{Periphery of the duct}} \quad (3)$$

If oil conductivity, σ is known, dielectric response of the oil insulation, $\epsilon_{oil}(\omega)$ can be obtained from equation (4). However, dielectric response of the pressboard insulation, $\epsilon_{PB}(\omega)$ has to be obtained by conducting FDS measurement on pressboard samples. Once individual responses are known, response of whole transformer insulation, $\epsilon_{tot}(\omega)$ can be estimated by the equation (5). Accordingly, the loss tangent of the whole insulation system, $\tan \delta_{tot}$ can be evaluated as in equation (6).

$$\epsilon_{oil}(\omega) = 2.2 - j \frac{\sigma(T)}{\epsilon_0 \omega} \quad (4)$$

$$\epsilon_{tot}(\omega) = \frac{1-Y}{\frac{1-X}{\epsilon_{oil}(\omega)} + \frac{X}{\epsilon_{PB}(\omega)}} + Y \times \epsilon_{PB}(\omega) \quad (5)$$

$$\tan(\delta)_{tot} = \frac{\epsilon''_{tot}(\omega)}{\epsilon'_{tot}(\omega)} \quad (6)$$

B. Principal Component Analysis (PCA)

The PCA is a data transformation technique that uses linear orthogonal transformation based on the variation of features for dimensionality reduction in pattern recognition [9, 10].

If the feature matrix is given by $X = [X_1 X_2 X_3 \dots X_n]$ where $X \in \mathbb{R}_{m \times n}$, first feature space is transformed to the origin of the feature space by shifting using the equation $\hat{X} = X - \bar{X}$, where $\bar{X}$ is the mean value of each feature calculated by the equation. Then the Covariance matrix is found as in equation (7). Afterwards Eigenvectors (V) and Eigenvalues (λ) of the matrix C are found. Then the feature space is transformed by mapping the feature vectors on to the Eigen vectors using the equation $Y = X \cdot V$, so that the new feature vectors are uncorrelated. By ranking the eigenvectors in order of their eigenvalues, highest to lowest, the PCs can be sorted in order of significance [9,10].

$$C = \frac{1}{n-1} \sum_{i=1}^n (X - \bar{X}) \cdot (X - \bar{X})^T \quad (7)$$

C. Support Vector Machine (SVM)

The SVMs are proved to be effective and promising techniques for data mining in a classification problem. The SVM algorithm defines a hyperplane that separates boundaries of different classes. Here, a multiclass classification is done by reducing it to a multiple binary classification problem and using a one-versus-one-strategy. For example, a classifier of 3 classes should need to have 3 binary classifiers [11].

A binary classifier is built by defining a hyper plane that separates the trained data into two categories. The hyper plane is defined by,

$$g(x) = wx + b = \begin{pmatrix} w_1 \\ w_2 \end{pmatrix} (x_1 \ x_2)^T + b = 0 \quad (8)$$

Where x represents features (i.e. principal component axis 1 and 2 in the proposed method), w is a vector of fitted linear coefficients, b is the bias scaler.

The hyperplane defines two sides of the space where one side's $g(x)$ value is positive and the other side's $g(x)$ is

negative. For the convenience of calculation, the whole space is scaled for all training data points [11].

$$y_t(w \cdot x_t + b) - 1 \geq 0 \quad (9)$$

To optimize the classification results, the larger the minimum distance, the better. In fact, maximizing the minimum distance is proved to be equal to minimizing $\frac{\|w\|^2}{2}$. In multidimensional space, the problem of finding the hyper plane remains the same [11]:

$$y = \begin{cases} \frac{\|w\|^2}{2}, & \text{objective function to minimize} \\ (w \cdot x_t + b) - 1 \geq 0, & \text{constraint function} \end{cases} \quad (10)$$

To solve (10), a Lagrange equation can be constructed and solved by deriving its partial derivative. Therefore, the hyper plane $g(x)$ is determined after w and b are computed. For a new data x_u , its type can be judged by the sign of $g(x_u)$ [11]. For a new test set, the final class is defined by the most votes given by the whole classifier inclusive of multiple binary classifiers.

III. METHODOLOGY

This study, considered as a pattern classification problem, has FDS results as the input and three moisture groups as the output. The data set consist of results obtained from field aged transformers and of results developed by XY model of laboratory prepared pressboard samples. The outputs groups were determined by moisture contents measured on those input samples. The test procedure is illustrated in Fig. 2.



Fig. 2. Overall test procedure

A. Preparation of Pressboard Samples

Fourteen number of disc shaped pressboard samples (diameter of 100 mm and thickness of 1.5 mm) were used to obtain dielectric response of pressboard insulation. Out of them, eight pressboard samples were treated with different moisture levels without aging whereas the rest of the samples were aged. First all the samples were dried inside an air circulation oven for 24 hrs at 65 °C followed by vacuum drying at 95°C for 24 hrs. Afterwards dry oil impregnated pressboard samples were prepared by impregnating with mineral oil. In order to prepare samples with different MCs, the impregnated samples were kept for 28 days inside different humidity control chambers having relative humidity levels of 5.5 %, 11.5%, 30.5%, 45.5% and 74.5% at 50 °C [12]. Another three sets of pressboard samples with an initial moisture contents of 2.2 %, 2.4 %, and 2.0 % were aged in three different hermetically sealed stainless-steel chambers. The aging process was performed at 120 °C with a power transformer proportion of copper conducting bars and core steel. The aging process was stopped at regular intervals of 28, 48, 62, and 84 days and samples were taken out for dielectric response analysis [13].

B Laboratory Modeled FDS Data

FDS measurements performed at 35 °C on the prepared pressboard samples were used to derive modelled FDS data for transformer insulations as explained in section II. Two X and Y parameters, each ranging from 0.15 to 0.3 and two oil conductivities ranging from 0.01 to 400 pS/m 35 °C were used to prepare 112 FDS data sets. The measured MCs of the 14 samples were in the range from 0.3% to 5.2%.

C Field Measured FDS Data

Twenty power transformers identified in different hydro power stations in Sri Lanka were selected for field measurements. Their service times varied from 15 to 40 years. Before taking the measurements, transformers were de-energized and kept for 3-4 days to reach equilibrium condition. The FDS measurements were then taken at each transformer at a peak voltage of 200 V and frequency range of 1 kHz - 0.1 MHz by using DIRANA insulation analyzer. The testing temperature was about 35 °C.

D Classification and Testing

In order to get a better training process, the loss tangent values corresponding to 22 frequency points in each data set were doubled by interpolation. Out of 132 data sets 80% to 20% data sets were selected for training and testing in a random basis. Accordingly, out of 105 data sets used for training had 89 and 16 data sets from laboratory modeled and field measured respectively. The rest of the 27 sets used for testing were divided as 21 and 6 for laboratory modeled and field measured data. During training, first the 44 PCs were estimated. Fig. 5 shows box plot of first 2 PCs. According to the figure it is evident that the first two PCs give higher class separation and therefore total dimensions were reduced to two for further analyses as the inputs of the classification problem. On the other hand, the outputs were three moisture classes. The three classes are dry (MC <1%), intermediate (MC from 1% to 3%) and wet (MC > 3%), indicating the condition of the transformer. The SVM technique was then used to estimate the moisture groups from input data.

IV. RESULTS AND DISCUSSION

Fig. 3 shows frequency dependency of loss tangent for some selected pressboard samples representing dry and intermediate states as well as non-aged and aged once. Results clearly indicate that the loss tangent values increase with the increase of both aging times as well as MC levels showing us the ambiguity of direct estimation of MC level of the transformer pressboard insulation when aging states are different. In addition, the modeled FDS plots produced by XY model representing the transformer insulation by adding transformer geometry and oil condition parameters on top of the pressboard moisture, this moisture estimation on transformer insulation based on overall FDS plot may become even tedious.

Fig. 4 shows FDS plots of three selected power transformers representing dry, intermediate, and wet conditions. Their service times are 13 years for dry, 27 years for intermediate, and 26 years for wet conditions respectively. The presented results are

in good agreement with the typical loss tangent characteristics of a power transformer. For example, with the increase of MC loss tangent values measured at very low frequency range increase, and also the loss peak shifts to right hand side with moisture increase. However, the typical shift of about -1 gradient line showing the variation of oil conductivity properties can also clearly be seen in the figure most probably due to the shift of loss peaks rather than oil conductivity increments. As moisture, aging, oil conductivity and temperature have similar effect of dielectric response, the FDS plots obtained from field aged transformers, provides us the difficulty of moisture estimation by direct observation.

Fig. 5 shows the box plots for the scores of PCs 1 and 2 according to moisture classes. This was used to identify the most useful PCs to visualize differences between moisture classes. As in here, the wet class box of PC1 does not overlap with any other boxes in different classes. Dry and Intermediate class boxes do have some differences in between them in both PC1 and PC2. The box plots of the rest of the PCs do not show any clear difference amongst moisture classes. Thus, PC1 is sufficient to separate dry and wet states of transformer insulation without using ML based approach. On the other hand, to get better separation, ML based SVM technique is required.

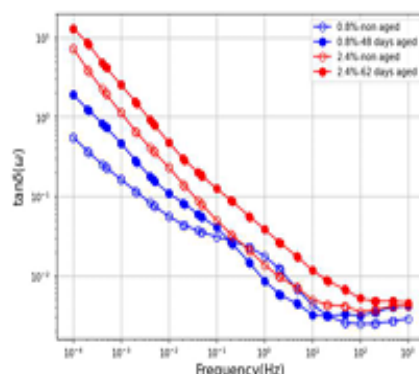


Fig. 3. FDS plots obtained from selected pressboard samples

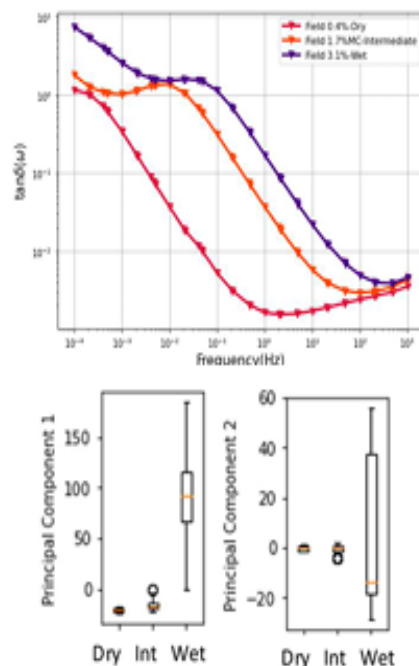


Fig. 5. Box plots of PC1 and PC2

Fig. 6 shows the 2D plot for trained samples based on PC1 and PC2. The figure clearly shows that PC1 for dry and intermediate samples are small and negative. On the other hand, PC1 for wet samples are positive and covers a larger spectrum. The PC1=0 boundary separates these two regions coarsely. Fig. 7 shows the 2D plot for trained samples based on PC1 and PC2. The separating boundaries obtained from SVM technique are also added. The results clearly indicate that the three moisture regions i.e. dry, intermediate and wet, are separated.

The test data set was plotted to the defined boundaries as shown in Fig. 8. Table 1 shows the confusion matrix (also known as error matrix) for the tested 27 samples. The accuracy of the trained model is 96 % (i.e. out of 27, only one sample was not correctly estimated). The incorrect sample results (remaining 4% accuracy) lies in the upper triangular elements of the confusion matrix. It was a dry sample indicated as an intermediate one and the result may not be harmful as far as the

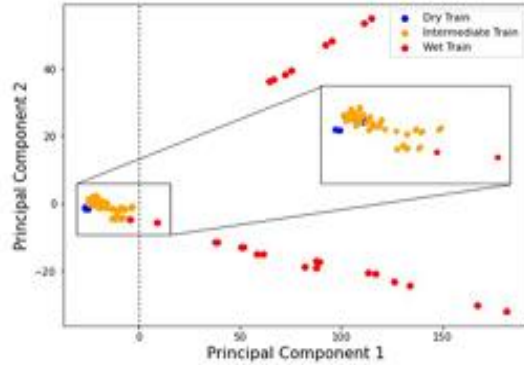


Fig. 6. Plot of trained data set for first two PCs

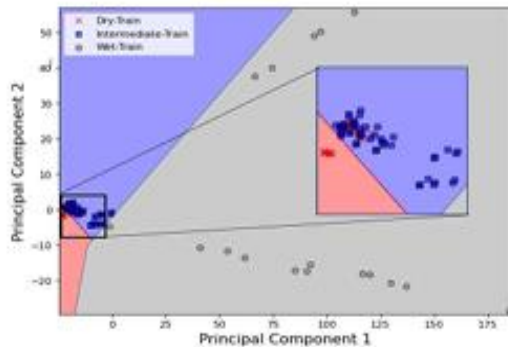


Fig. 7. First two principal components with moisture separation boundaries for the trained data set

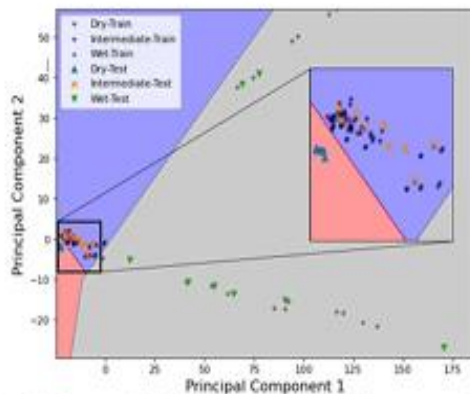


Fig. 8. First two principal components with moisture separation boundaries for trained and test data set

Table 1 Confusion matrix

Actual	Estimated cases		
	Dry	Intermediate	Wet
Dry	8	1	0
Intermediate	0	9	0
Wet	0	0	9

condition assessment process is concerned compared to its wise versa. By considering all, it is evident that the proposed method is a simple and promising method for moisture estimation in a power transformer.

V. CONCLUSIONS

The Principal Components mapped from the dielectric spectroscopy of transformer insulation measured from 0.1 mHz to 1 kHz frequency range indicate that the first two components are sensitive to moisture content. First components itself is sufficient to identify insulation having more than 3% moisture content. The SVM based ML approach applied on two principal components allows to classify insulation states as dry ($MC < 1\%$), intermediate ($1\% < MC < 3\%$) and wet ($MC > 3\%$). The tested samples verify the accuracy of the trained set up to 96%. It can be concluded that principal component analysis together with support vector machine approach can be used to assess the condition of power transformer insulation based on its measured FDS plots.

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